L-2/T-II/CSE **Date: 21/05/2026**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-II B. Sc. Engineering Examinations 2023-2024

Sub: **MATH 243** (Probability and Statistics)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

Symbols used have their usual meaning

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) Can two events be mutually exclusive and independent simultaneously? Justify your answer. **(6)**

(b) Define multiplication law of probability. Suppose that we have two boxes A and B. The first box contains 5 white balls and 7 red balls, while the second box contains 3 white balls and 8 red balls. One ball is transferred from the first box to the second box unseen and then a ball is drawn from the second box. What is the probability that this ball is red? **(12)**

(c) An opinion poll in city shows that 45 percent of the city dwellers support party A, 40 percent support party B and remaining 15 percent support other parties C. Previous records reveal that in city elections, 65 percent of A, 80 percent of B and 50 percent of the other party supporters turned up to cast their votes. A person in the city is chosen at random and it sis learned that he did not cast his vote in the last election. What is the probability that he is a supporter of A? B? C? **(17)**

2. (a) The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable X with cumulative distribution function

$$F(x)=\begin{cases}1-e^{-8x}, & x\geq 0\\ 0, & \text{otherwise}\end{cases}.$$ **(15)**

Find the probability of waiting less than 12 minutes between successive speeders; using the cumulative distribution function of X and the probability density function of X.

(b) From a bag of pens containing three red pens, two blue pens and three black pens, a random sample of two pens is selected. If X is the number of red pens, and Y is the number of blue pens in the sample, then work on the following: **(20)**

(i) Plot $f(x,y)$ verses (x,y), where $f(x,y)$ denotes joint distribution.

(ii) $P[(X,Y)\in A]$, where $A=\{(x,y)$ such that $x+y\leq 1\}$.

(iii) Marginal distributions of X and Y.

(iv) Variance of X.

(v) Covariance between X and Y.

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3. (a) State Chebyshev's theorem. Seventy new jobs are opening up at an automobile manufacturing plant, and 1000 applicants show up for the 70 positions. To select the best 70 from among the applicants, the company gives a test that covers mechanical skill, manual dexterity, and mathematical ability. The mean grade on this test turns out to be 60, and the scores have a standard deviation of 6. Can a person who scores 84 count on getting one of the jobs? Assume that the distribution is symmetric about the mean. **(11)**

(b) Establish the relation between binomial and Poisson distribution. Suppose that, telephone calls arrive at a switchboard at a mean rate of 0.5 calls per minutes. Calculate the probability that two calls will arrive in a particular five minute period. **(12)**

(c) The probability that a person recovers from a rare blood disease is 0.4, if 15 people are known to have contacted this disease, what is the probability that **(12)**

(i) at least 10 people survive?

(ii) from 3 to 8 survive?

(iii) exactly 5 people die?

4. (a) Show that the mode of normal distribution is equal to μ, where μ is the mean of that distribution. **(11)**

The Phillips Bangladesh manufactures electric bulbs that have a length of life that is normally distributed with mean equal to 800 hours and standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours. (Necessary table is attached)

(b) What is Pearson's correlation coefficient? How do you modify your formula to compute the rank correlation coefficient when the observations are repeated? Calculate the rank correlation coefficient for the following data on hourly sales (x) and expenses (y) of 10 stores: **(12)**

Sales (x)	50	50	55	60	65	65	65	60	60	60
Expenses (y)	11	13	14	16	16	15	15	14	13	13

(c) Briefly mention the objectives of correlation and regression analysis. A test was made to different doses of nitrogen (N) on rice yield. The doses of phosphorus (P) and potassium (K) were kept at the optimum level for proper growth of plant. The following data were recorded. **(12)**

Nitrogen doses (lb/acre)	0	1	2	3	4	5
Yield (maunds/acre)	15	25	40	55	52	43

Fit a second degree polynomial to the data and obtain coefficient of determination for the data.

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) What do you mean by Stem-and-Leaf plots? How to create Stem-and-Leaf plots? Explain with examples. **(10)**

(b) Explain the scatter plot with an example briefly. Prices (in dollars) and the ratings (out of 5 stars) of 5 different restaurants are recorded and plotted in a scatter plot. Interpret the scatter plot to determine if there is any relationship between price and rating. **(13)**

Restaurant	A	B	C	D	E
Prices	22	29	15	18	23
Ratings	3.5	4.0	3.0	4.0	3.5

(c) The mean and standard deviation of 200 items are found to be 60 and 20, respectively. If at the time of calculation, two items are wrongly taken as 3 and 67 instead of 13 and 17, find the correct mean and standard deviation. What is the correct coefficient of variation? **(12)**

6. (a) Calculate the first four moments about the mean and also find the value of β_1 and β_2 from the following data. Hence, comment on the nature of the distribution. **(13)**

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	8	12	20	30	15	10	5

(b) What do you mean by the central limit theorem in statistics? Write an algorithm to solve problems involving the central limit theorem. Discuss its applications in Computer Science. **(10)**

(c) Explain with an example, stratified random sampling. An online medicine shop claims that the mean delivery time for medicines is less than 120 minutes with a standard deviation of 30 minutes. Is there enough evidence to support this claim at a 0.05 significance level if 49 orders were examined with a mean of 100 minutes? Given that $Z = -1.645$ at 5% level of significance. **(12)**

7. (a) Write short notes on the goodness of fit, one-tailed and two-tailed test, Degrees of freedom, and null-hypothesis and alternative hypothesis. **(10)**

(b) A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved at 1% level of significance? Given that at 1% level of significance, $Z = \pm 2.58$. **(12)**

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(c) Two independent samples of 8 and 7 items, respectively, had the following values of the variable (weight in ounces). Is the difference between the means of the sample significant? Use 5% level of significance. Note that for $\upsilon = 13$, $t_{0.05} = 2.16$. **(13)**

Sample I	9	11	13	11	15	9	12	14
Sample II	10	12	10	14	9	8	10	

8. (a) A random sample of 200 employees was drawn from a business enterprise to see if there is a relationship between educational attainment and job performance. From the following data, do you find any relationship between educational attainment and the job performance of these employees? Test at 5% level of significance, where $\chi^2_{0.05,4} = 9.49$. **(15)**

Job performance	Educational attainment			
	Below Primary	College	University	Total
Excellent	10	40	10	60
Good	30	30	20	80
Fair	10	30	20	60
Total	50	100	50	200

(b) The following table gives the retail prices of a commodity in some shops selected in their big cities. Carry out the analysis of variance to determine whether there is a significant difference in the prices charged in these cities at the 5% level of significance, with $\upsilon_1 = 2$ and $\upsilon_2 = 9$, and $F_{0.05} = 4.26$. **(20)**

City	Price per unit (TK)			
A	15	18	23	24
B	23	19	18	24
C	19	16	22	31

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$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du$$

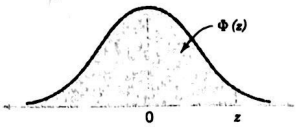


TABLE A.1 Cumulative Standard Normal Distribution (Continued)

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.500000	0.503989	0.507978	0.511967	0.515953	0.519939	0.532922	0.527903	0.531881	0.535856
0.1	0.539828	0.543795	0.547758	0.551717	0.555760	0.559618	0.563559	0.567495	0.571424	0.575345
0.2	0.579260	0.583166	0.587064	0.590954	0.594835	0.598706	0.602568	0.606420	0.610261	0.614092
0.3	0.617911	0.621719	0.625516	0.629300	0.633072	0.636831	0.640576	0.644309	0.648027	0.651732
0.4	0.655422	0.659097	0.662757	0.666402	0.670031	0.673645	0.677242	0.680822	0.684386	0.687933
0.5	0.691462	0.694974	0.698468	0.701944	0.705401	0.708840	0.712260	0.715661	0.719043	0.722405
0.6	0.725747	0.729069	0.732371	0.735653	0.738914	0.742154	0.745373	0.748571	0.751748	0.754903
0.7	0.758036	0.761148	0.764238	0.767305	0.770350	0.773373	0.776373	0.779350	0.782305	0.785236
0.8	0.788145	0.791030	0.793892	0.796731	0.799546	0.802338	0.805106	0.807850	0.810570	0.813267
0.9	0.815940	0.818589	0.821214	0.823815	0.826391	0.828944	0.831472	0.833977	0.836457	0.838913
1.0	0.841345	0.843752	0.846136	0.848495	0.850830	0.853141	0.855428	0.857690	0.859929	0.862143
1.1	0.864334	0.866500	0.868643	0.870762	0.872857	0.874928	0.876976	0.878999	0.881000	0.882977
1.2	0.884930	0.886860	0.888767	0.890651	0.892512	0.894350	0.896165	0.897958	0.899727	0.901475
1.3	0.903199	0.904902	0.906582	0.908241	0.909877	0.911492	0.913085	0.914657	0.916207	0.917736
1.4	0.919243	0.920730	0.922196	0.923641	0.925066	0.926471	0.927855	0.929219	0.930563	0.931888
1.5	0.933193	0.934478	0.935744	0.936992	0.938220	0.939429	0.940620	0.941792	0.942947	0.944083
1.6	0.945201	0.946301	0.947384	0.948449	0.949497	0.950529	0.951543	0.952540	0.953521	0.954486
1.7	0.955435	0.956367	0.957284	0.958185	0.959071	0.959941	0.960796	0.961636	0.962462	0.963273
1.8	0.964070	0.964852	0.965621	0.966375	0.967116	0.967843	0.968557	0.969258	0.969946	0.970621
1.9	0.971283	0.971933	0.972571	0.973197	0.973810	0.974412	0.975002	0.975581	0.976148	0.976705
2.0	0.977250	0.977784	0.978308	0.978822	0.979325	0.979818	0.980301	0.980774	0.981237	0.981691
2.1	0.982136	0.982571	0.982997	0.983414	0.983823	0.984222	0.984614	0.984997	0.985371	0.985738
2.2	0.986097	0.986447	0.986791	0.987126	0.987455	0.987776	0.988089	0.988396	0.988696	0.988989
2.3	0.989276	0.989556	0.989830	0.990097	0.990358	0.990613	0.990863	0.991106	0.991344	0.991576
2.4	0.991802	0.992024	0.992240	0.992451	0.992656	0.992857	0.993053	0.993244	0.993431	0.993613
2.5	0.993790	0.993963	0.994132	0.994297	0.994457	0.994614	0.994766	0.994915	0.995060	0.995201
2.6	0.995339	0.995473	0.995604	0.995731	0.995855	0.995975	0.996093	0.996207	0.996319	0.996427
2.7	0.996533	0.996636	0.996736	0.996833	0.996928	0.997020	0.997110	0.997197	0.997282	0.997365
2.8	0.997445	0.997523	0.997599	0.997673	0.997744	0.997814	0.997882	0.997948	0.998012	0.998074
2.9	0.998134	0.998193	0.998250	0.998305	0.998359	0.998411	0.998462	0.998511	0.998559	0.998605
3.0	0.998650	0.998694	0.998736	0.998777	0.998817	0.998856	0.998893	0.998930	0.998965	0.998999
3.1	0.999032	0.999065	0.999096	0.999126	0.999155	0.999184	0.999211	0.999238	0.999264	0.999289
3.2	0.999313	0.999336	0.999359	0.999381	0.999402	0.999423	0.999443	0.999462	0.999481	0.999499
3.3	0.999517	0.999533	0.999550	0.999566	0.999581	0.999596	0.999610	0.999624	0.999638	0.999650
3.4	0.999663	0.999675	0.999687	0.999698	0.999709	0.999720	0.999730	0.999740	0.999749	0.999758
3.5	0.999767	0.999776	0.999784	0.999792	0.999800	0.999807	0.999815	0.999821	0.999828	0.999835
3.6	0.999841	0.999847	0.999853	0.999858	0.999864	0.999869	0.999874	0.999879	0.999883	0.999888
3.7	0.999892	0.999896	0.999900	0.999904	0.999908	0.999912	0.999915	0.999918	0.999922	0.999925
3.8	0.999928	0.999931	0.999933	0.999936	0.999938	0.999941	0.999943	0.999946	0.999948	0.999950
3.9	0.999952	0.999954	0.999956	0.999958	0.999959	0.999961	0.999963	0.999964	0.999966	0.999967

Table for Q 4(a)

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